

Clash Royale simulator: current handoff

Evidence-based Python simulator status, updated 20 August 2026. Scope: simulator only; no emulator or live-bot behavior was changed.

1. Current position

Public card catalogue	120 RoyaleAPI public cards; every one maps uniquely to shipped client data
Random deck viewer	119 resolvable public cards sampled into unique 8-card decks; Party Rocket excluded because its graph is quarantined
Local client data	207 card rows parsed; 176 deployable units clear in the top-level raw-field scan
Spells	27 resolvable spell actions; zero unresolved public spell rows
Remaining source graphs	9, all named and explicitly blocked on live collision or trajectory calibration
RL readiness	NOT READY - controlled live geometry/contact/projectile evidence is still required
Tests	1308 passing plus 2 pinned gaps; board symmetry, elixir rates, determinism, speed units and parser anchoring now pinned
Fixed this shift	Mirror resolved cards at level 12, where all 37 verified combat_rules overrides stopped applying - a mirrored Evolved Witch had 922 hitpoints against a verified 1451. Verified values are now carried along the client curve and are exact at the level they were verified at
Capture plan	python -m sim.probe_plan prints 18 controlled clips, each paired with the engine's current prediction
Known measurement hazard	self-play A/B is seat-confounded: BrainPolicy wins 60.5% from the bottom seat (z=+3.69) though the board is provably symmetric

The simulator is approximately 80% complete for a credible RL environment. The remaining risk is shared game physics and timing, not basic card-stat coverage.

2. What is implemented

The simulator reads shipped Supercell CSV/TOML data and a versioned RoyaleAPI card snapshot. It covers charge, invulnerable dashes, shields, splash, projectile flight, death damage/spawns, periodic spawners, timed buffs, knockback, burrowing, tower windup, pathfinding, spells, evolutions, champions and current hero action handlers.

The viewer now supports **--random-decks**. It builds two random public-card decks and uses a generic legal-placement policy. That policy now presses an engaged champion/hero ability when it is legal, so Golden Knight's paid dash is exercised instead of being silently ignored.

Viewer command

```
.\venvs\buildabot\Scripts\python.exe -m sim.watch --random-decks --seed 8 --speed 2
```

Seed 8 includes Golden Knight in the bottom deck. The dash occurs after deployment, engagement and sufficient ability elixir.

3. Evidence and verification added

Nineteen user-supplied MuMu gameplay recordings were catalogued with source paths, dimensions, frame counts and SHA-256 hashes. All hashes currently verify. The clips are approximately 30 fps with overlapping play, so they remain contextual evidence only - not calibration proof.

The calibration workflow now includes: recording inventory; motion-event triage; exact frame export with source hash provenance; source-file hash verification; and a strict readiness gate. Accepted evidence must cite a controlled 50+ fps capture, valid source frames, card levels/deployment, observed outcome and a real pytest regression test.

Recent checks

Core simulator/viewer/deck/ability regression suite: passing. Random-deck visual smoke run: passing. Recording hash verification: 19/19 passing. Catalogue audit: 120/120 public cards mapped. Run **python -m sim.readiness** for the current strict training gate.

4. What must be done next

Do not fill these gaps with guessed sprite boxes, collision constants or projectile paths. Capture isolated 60 fps controlled clips, then calibrate and regression-test each result.

Shared calibration matrix

Map anchors	Tower centres, river, bridges and legal deployment boundary
Troop contact	Same-size/mixed-size contacts, including King Tower and open-lane cases
Building contact	Routes and contact around Cannon, Tesla and Goblin Cage
Projectile timing	Moving/stationary targets for homing and non-homing shots
Spell timing	Fireball, Arrows, Zap and Tornado cast-to-impact/miss behavior

Nine named mechanics also remain calibration-gated

Executioner Evolution returning axe; Goblin Drill Evolution emergence; Princess Evolution trail; Hero Balloon payload; Hero Magic Archer teleport/triple shot; Firecracker fan/deflection; Hero Mega Minion warp; Monk non-homing and spell reflection; Hero Wizard tornado pull.

5. Handoff commands

```
python -m sim.coverage - current card/spell coverage
```

```
python -m sim.action_audit - exact remaining source-graph calibration gates
```

```
python -m sim.readiness - strict RL training readiness
```

```
scripts/verify_live_probe_assets.py - verify recorded-video hashes
```

Expected remaining effort: 1-3 focused engineering days after measurements for deterministic cleanup, plus 1-3 weeks for credible collision, placement, interception and timing parity. Training should remain blocked until the probe matrix and the nine gates are evidenced and tested.